GUIDE FOR SAT EXAM

Mechanical Properties of Solids and Circular Motion Questions for SAT Exam

VSAQ

Very Short Answer Questions · 2 Marks

1 **A copper wire of 1 mm diameter is stretched by applying a force of 10 N. Find the stress in the wire.**

A: Given diameter $d = 1\,\text{mm} \Rightarrow$ radius $r = \frac{1}{2}\,\text{mm} = \frac{1}{2} \times 10^{-3}\,\text{m} = \frac{10^{-3}}{2}\,\text{m}$. Also force $F = 10\,\text{N}$

$$\therefore \text{Stress } \frac{F}{A} = \frac{F}{\pi r^2} = \frac{10}{\frac{22}{7} \times \left(\frac{10^{-3}}{2}\right)^2} = \frac{10}{\frac{22}{7} \times \left(\frac{10^{-6}}{4}\right)} = \frac{140}{11} \times 10^6 = 1.273 \times 10^7\ \text{Nm}^{-2}$$

2 **A tungsten wire of length 20 cm is stretched by 0.1 cm. Find the strain on the wire.**

A: Given $l = 20\,\text{cm}$, $e = 0.1\,\text{cm}$

Formula : $\text{Strain} = \frac{e}{l} = \frac{0.1}{20} = 0.005$

3 **State Hooke's law of elasticity.**

BMP

A: Hooke's Law : Within the elastic limit, the stress is directly proportional to strain.

Stress $\propto$ Strain $\Rightarrow$ Stress $= E$ (Strain). Here E is called as modulus of elasticity.

4 **What is the order of bulk modulus in the three states of matter?**

A: $B_{\text{Solids}} > B_{\text{Liquids}} > B_{\text{Gases}}$

i) Solids are **least compressible.**

ii) Liquids are moderately **compressible.**

iii) Gases are highly **compressible** (about **one million times** more compressible than solids)

5 **Define compressibility. Mention its equation.**

A: Compressibility is defined as the reciprocal of bulk modulus. It is the fractional change in volume per unit increase in pressure and denoted by k.

Formula : $k = \frac{1}{B} = -\frac{1}{\Delta p} \times \frac{\Delta V}{V}$

SI unit: Pa^{-1} or $N^{-1} m^2$

A material with higher bulk modulus has lower compressibility.

6 **What is Poisson's ratio? State the Poisson ratio limits of steel.**

A: Poisson's ratio (σ) is defined as the ratio of lateral strain to longitudinal strain within the elastic limit.

Formula : $\sigma = \frac{\text{Lateral strain}}{\text{Longitudinal strain}} = \frac{\Delta d/d}{\Delta L/L}$

For Steel, Poisson's ratio lies between **0.27 and 0.30**

7 **Two identical solid balls, one of ivory and the other of wet clay, are dropped from the same height onto the floor. Which one will rise to a greater height after striking the floor? Why?**

A: Ivory is more elastic than wet-clay. So ivory ball shall rise to a greater height as compared to a clay ball. In fact, the ball of wet-clay may not rise at all.

SAQ Short Answer Questions

8 **Explain the concept of Elastic potential energy in a stretched wire and hence obtain the expression for it.**

BMP

A: When a wire is stretched by an external force, work is done against the internal restoring forces of the material. This work is stored in the wire as **elastic potential energy**.

Consider a wire of **length L** and **cross-sectional area A** suspended from a rigid support.

Let a force **F** be applied at its free end, producing an extension **e**.

Initially, the force acting on the wire is zero, and it gradually increases to **F**.

Therefore, the **average force** acting during the extension is $F_{avg} = \frac{0 + F}{2} = \frac{F}{2}$

The extension produced in the wire is **e**.

Hence, the work done in stretching the wire is $W = F_{avg} \times e = \frac{F}{2} \times e = \frac{1}{2} Fe$

This work is stored in the wire as **elastic potential energy**.

$\therefore U = \frac{1}{2}Fe$. Here $U =$ Elastic potential energy, $F =$ Applied force, $e =$ Extension produced

Elastic Potential Energy per Unit Volume:

Volume of the wire $V = AL$

$\therefore \frac{U}{V} = \frac{\frac{1}{2}Fe}{AL} = \frac{1}{2}\left(\frac{F}{A}\right)\left(\frac{e}{L}\right)$ $\quad\left[\because \frac{F}{A} = \text{Stress};\ \ \frac{e}{L} = \text{Strain}\right]$

Elastic potential energy per unit volume $= \frac{1}{2} \times$ Stress $\times$ Strain

This work is stored in the wire in the form of elastic potential energy (U).

9 Under what condition will a car skid on a leveled circular road?

A: A car will skid on a **leveled circular road** when the **frictional force** between the tires and the road is no longer sufficient to provide the required **centripetal force** for the turn.

The car will skid if the required centripetal force exceeds this limit of maximum available static friction.

$$\text{Centripetal force } f = \frac{mv^2}{R} \tag{1}$$

At maximum velocity the friction reaches its limiting value

$$f_{\text{max}} = \mu_s N = \mu_s mg \tag{2}$$

From (1) and (2),

$$\frac{\cancel{m}v^2}{R} = \mu_s(\cancel{m}g) \Rightarrow \frac{v^2}{R} = \mu_s(g) \Rightarrow v^2 = \mu_s Rg \qquad \text{Hence, } v_{\text{max}} = \sqrt{\mu_s Rg}$$

If $v > \sqrt{\mu_s Rg}$ then the car skids.

If $v \leq \sqrt{\mu_s Rg}$ then the car does not skid.

10 Briefly explain centrifugal force with suitable examples.

A: Centrifugal force is a pseudo force that appears to act radially outward on a body moving in a circular path. It is felt only in a rotating (non-inertial) frame of reference attached to the body.

Ex:

(i) A passenger sitting in a car taking a sharp turn feels thrown outward, away from the centre of the turn, as if pushed by a centrifugal force.

(ii) Mud flies off tangentially (outward, in the rotating frame's perspective) from a rapidly spinning bicycle wheel.

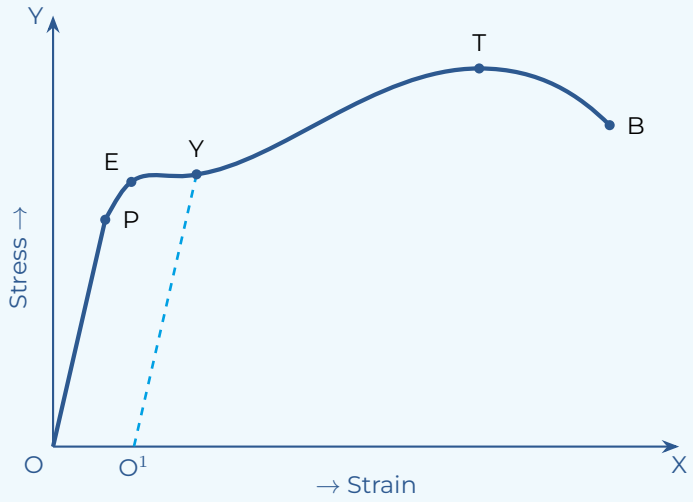
(iii) In a merry-go-round, a person feels pushed outward, even though the only real force acting is the inward tension in the supports providing centripetal force.

LAQ Long Answer Questions

11 **Describe the behaviour of a wire under gradually increasing load.**

TS 15,17,18,20,22,24,25

A: Consider a wire suspended from a rigid support and loaded at the other end. Suppose the load is increased gradually until it breaks.
A graph is plotted between strain on the X-axis and the stress on the Y-axis.
The nature of graph is shown here.



Behaviour of a wire under increasing load:

1) Proportionality limit(OP): The part OP is a straight line which shows that stress is proportional to strain. The wire obeys Hooke's law up to the point P. So P is called the proportionality limit of the wire.

2) Elastic limit (PE): Beyond P up to E, the graph is slightly curved. When the load is removed, the wire will regain its natural length. Up to E, the wire can be deformed elastically.

3) Yielding point(Y): Beyond elastic limit, when the load is removed at the point Y, the wire does not regain its natural length completely. It will have a permanent increase in length. In the region EY the wire shows plastic behaviour.

4) Tensile Point(T): Beyond the point Y, the strain increases rapidly without any increase in the load. Even if the load is not removed, the strain increases continuously till the wire reaches the point T. The stress corresponding to T is called the tensile strength of the given material.

5) Breaking Point(B): Beyond the point T, the wire shows necks at few points along the length of the wire. Consequently, the wire breaks at B. This point B is called 'breaking point'.